

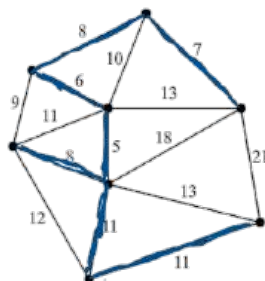
Network Concepts Topic Test 1

Solution

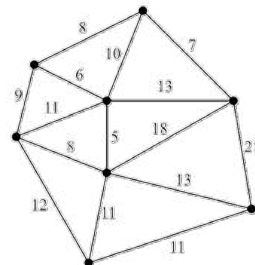
Total Marks: 30

Time: 50 minutes

- 1 What is the length of the minimum spanning tree in the network below?

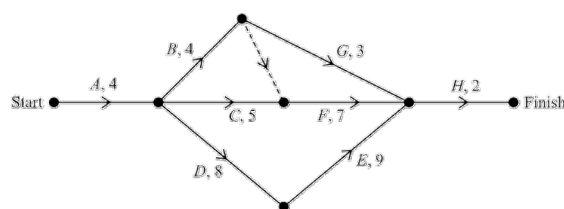


$$\text{length} = 5 + 6 + 7 + 8 + 8 + 11 + 11 = 56$$



2

- 2 The network below shows the activities that need to occur for a project to be completed. Complete the activity table below for the network shown.



3

Activity	Duration	Immediate predecessor
A	4	—
B	4	A
C	5	A
D	8	A
E	9	D
F	7	B, C
G	3	B
H	2	E, F, G

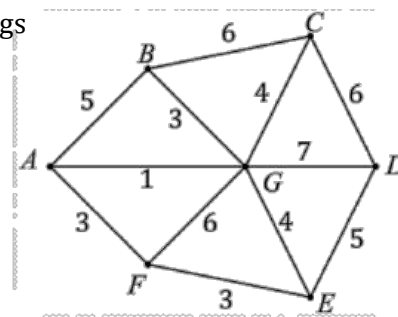
- 3 The diagram shows a network. What is the sum of the degrees of all the vertices in this network?

12



1

- 4 The network diagram shows the walking path between buildings in a school. The average time (in minutes) to walk between buildings is shown on the edges of the diagram. Determine which route has the shortest average time to walk from building B to F. Justify your answer with calculations.



2

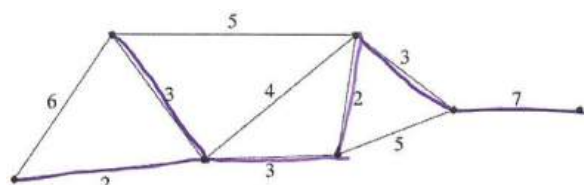
The shortest path is B - G - A - F

$$3 + 1 + 3 = 7 \text{ min}$$

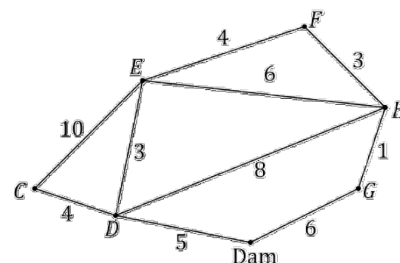
The shortest average walk time is 7 minutes.

- 5 The diagram represents a network with weighted edges.

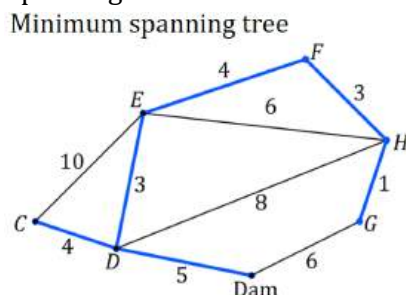
What is the total length of the minimum spanning tree?



- 6 The network diagram below shows pipes providing water from a dam to six farms. These pipes connect the farms to each other and to the dam. The lengths of the pipes (in km) are shown. A plumber is hired to check the water pipes.

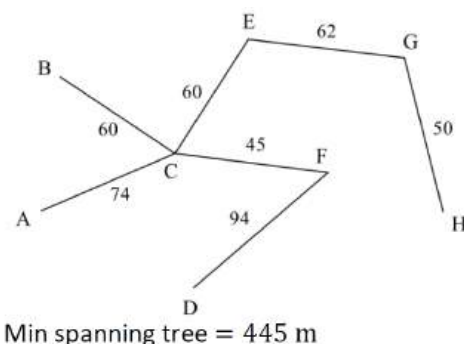


- a. Draw a minimum spanning tree for the network diagram. 2



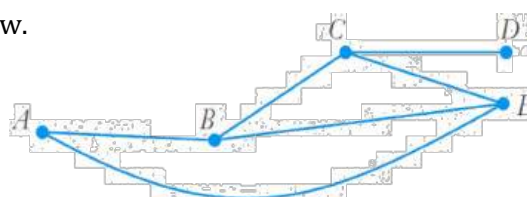
- b. The system of pipes is due for replacement. What is the minimum length of pipe required to service all of the farms? 1
Length = $4 + 5 + 3 + 4 + 3 + 1 = 20$ km

- 7 The weighted network below models bird-watching platforms (vertices) within a national park that are connected together by footpaths (edges). The weightings on the edges of the graph indicate the distance in metres between the bird watching platforms. Draw a minimum spanning tree for this network and determine its length. 3



- 8 Five towns are connected by the network diagram below.

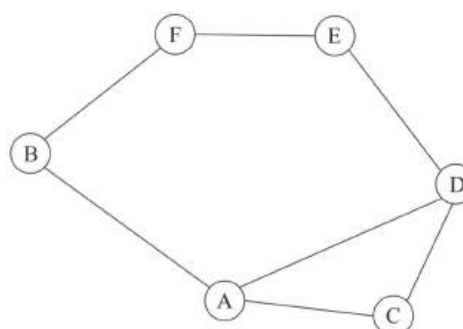
- a. State the vertex with an even degree. A
- b. Show that the sum of degrees is equal to twice the number of edges.



Edges = 6 ✓
Degree = $2 + 3 + 3 + 1 + 3 = 12$

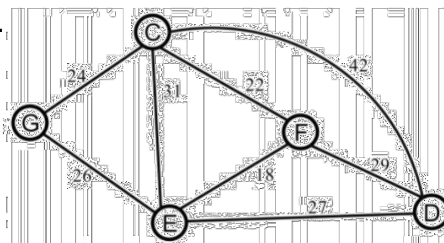
- 9 State a trail for the following network. 1

ABFEDCAD
OR
ABFEQACD

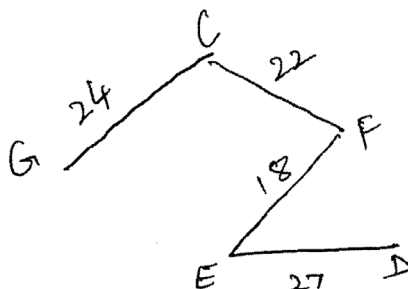


- 10 There are five towns that are connected by a series of roads.

The distance between each town is shown on the network diagram below:



- a. A funding program is introduced to re-seal some of the roads. Draw a minimum spanning tree so that it is possible to travel between any two towns (not necessarily directly) along a newly sealed road. 2

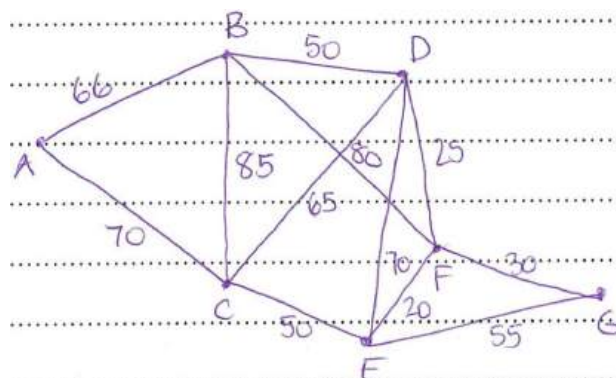


- b. What is the minimum length of road that would need to be re-sealed? 1

$$24 + 22 + 18 + 27 = 91$$

- 11 The table below shows the distances of the roads linking 7 towns on Stephen's Island, all distances are in kilometres.

	A	B	C	D	E	F	G
A	—	66	70	—	—	—	—
B	66	—	85	50	—	80	—
C	70	85	—	65	50	—	—
D	—	50	65	—	70	25	—
E	—	—	50	70	—	20	55
F	—	80	—	25	20	—	30
G	—	—	—	—	55	30	—



- a. Draw a weighted network to represent the roads that link the towns. 2

- b. What is the shortest path, and its length, from town A to town G? 2

$$A B D F G = 171$$

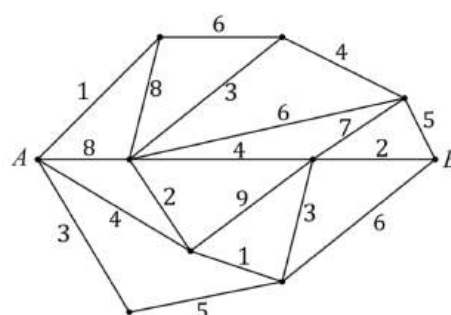
$$A B F G = 176$$

$$A C E F G = 170$$

$$A C E F G = 170 \text{ km}$$

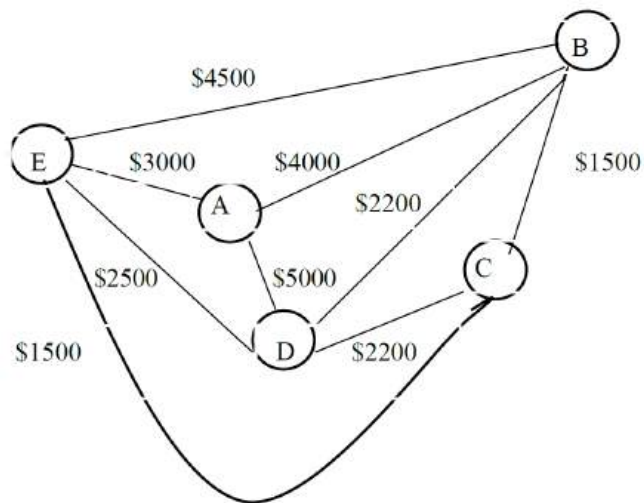
- 12 The network diagram below shows the distances, in kilometres, along a series of roads that connect town A to town B. What is the shortest distance, in kilometres, from town A to town B?

10



- 13 A spanning tree contains four edges. How many vertices are in the network? 1
 A. 2 B. 3 C. 4 D. 5 (D) 5
- 14 The costs (in \$) of connecting various locations on a school campus with computer cable are given in the table below. A blank space indicates no direct connection. Draw a network to represent this situation, showing the cost of connection along each arc. 2

	A	B	C	D	E
A	—	4000		5000	3000
B	—	—	1500	2200	4500
C	—	—	—	2200	1500
D	—	—	—	—	2500



Network Concepts Topic Test 2

Solution

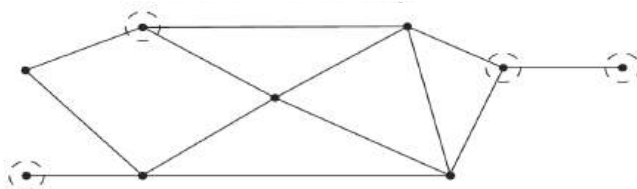
Total Marks: 30

Time: 50 minutes

- 1 A network diagram is shown.

How many vertices have odd degrees?

There are four vertices of odd degree.



- 2 Consider the network diagram below. Which of the following is a cycle?

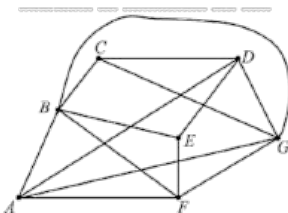
A. ABEF

B. ABCDEFG

D. FEDGCBF

C. ABCDGAB

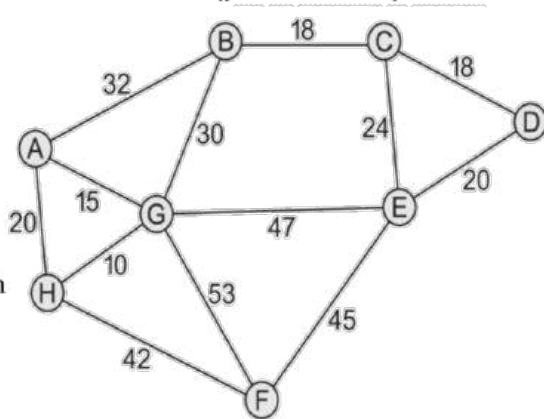
D. FEDGCBF



- 3 The above network shows connecting paths for a cable network. It costs \$65 per metre to lay the cable. Use a minimum spanning tree to determine the least cost to connect this network.

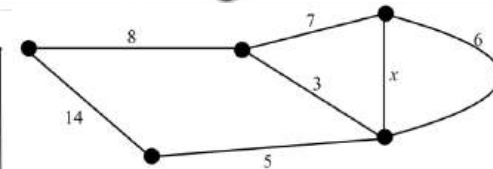
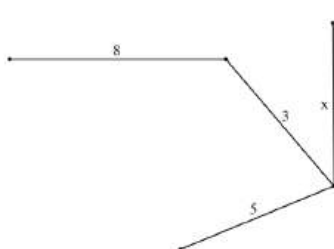
Minimum spanning tree : $15 + 10 + 42 + 30 + 18 + 18 + 20 = 153$ m

Cost : $153 \text{ m} \times \$65 = \9945



- 4 The minimum spanning tree for the below network includes the edge x .

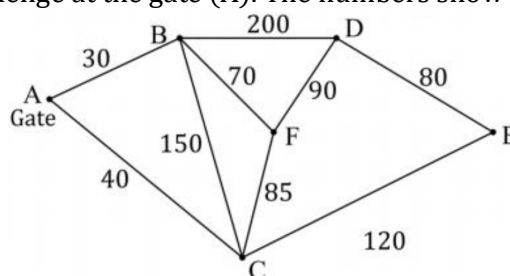
- a. Draw the minimum spanning tree for the above network.



x must be less than 6

- b. If x is included in the minimum spanning tree, what can you say about the value of x ?

- 5 The network diagram below shows an orienteering challenge with five tasks (B, C, D, E, F) that are joined by paths. You must start and finish the challenge at the gate (A). The numbers show the distance along the paths in metres. By first stating the path, calculate the minimum distance a team would have to run to complete every task in the challenge, starting and ending at the gate.



- 6 The path is **ABFDECA**

$$30 + 70 + 90 + 80 + 120 + 40 = 430 \text{ m}$$

tations for the school's annual

fundraising event. The table shows the distances (in metres) between the seven activity stations.

	A	B	C	D	E	F	G
A	–	66	70	–	–	–	–
B	66	–	85	50	–	80	–
C	70	85	–	65	50	–	–
D	–	50	65	–	70	25	–
E	–	–	50	70	–	20	55
F	–	80	–	25	20	–	30
G	–	–	–	–	55	30	–

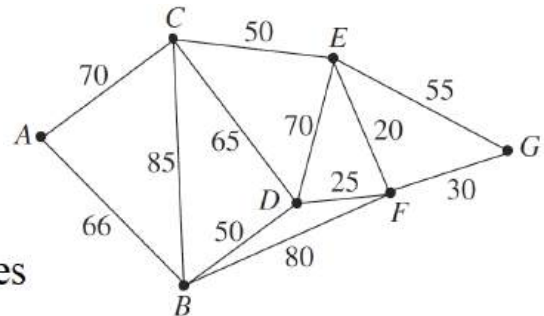
- a. Using the data from the table above, draw a weighted network diagram in the space below.

- b. Determine the shortest path from activity station A to activity station G.

shortest path = *ACEFG*

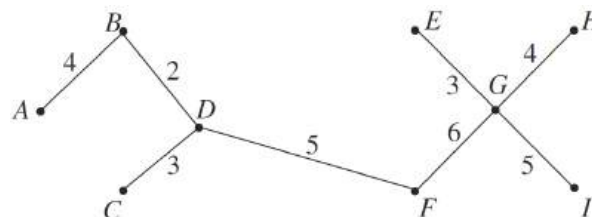
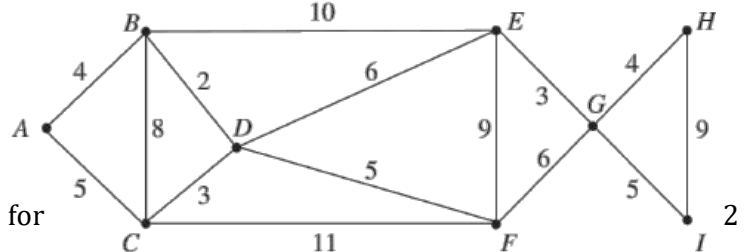
- c. What is the weight of the shortest path?

weight = $70 + 50 + 20 + 30 = 170$ metres



- 7 The network diagram shows the cost, in hundreds of dollars, to connect electrical cables to multiple points in a building.

- a. Draw the minimum spanning tree for this network in the space below.

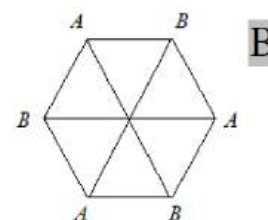
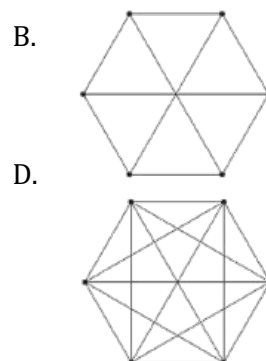
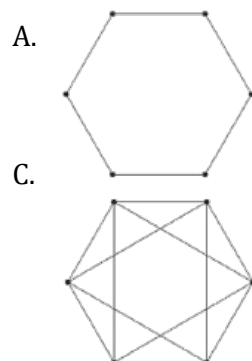


- b. Using the minimum spanning tree, calculate the minimum cost to connect a cable to every point in the building.

$$4 + 2 + 3 + 5 + 6 + 3 + 4 + 5 = 32$$

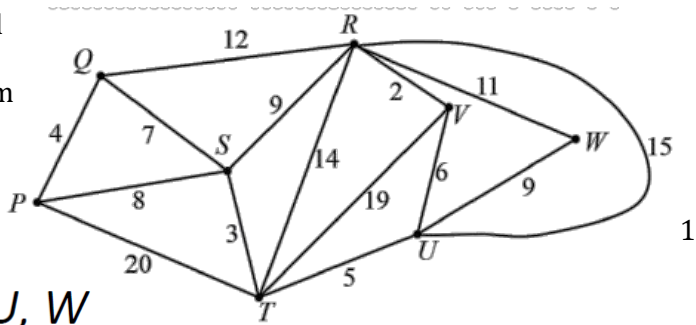
The network diagram shows the cost in hundreds of dollars. Therefore, the minimum cost is \$3200.

- 8 Team A and Team B have entered a chess competition. Team A and Team B have three members each. Each member of Team A must play each member of Team B once. Which of the following network diagrams could represent the chess games to be played?

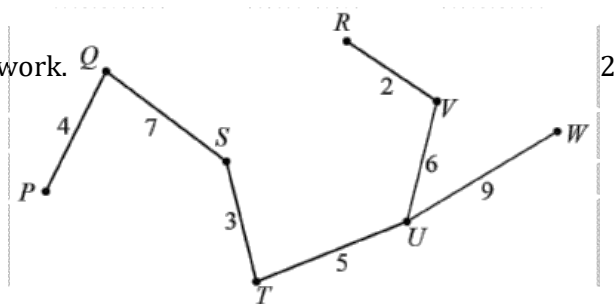


B

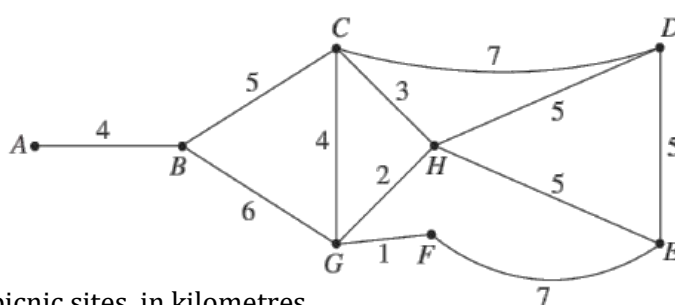
- 9 The graph of a university grounds is pictured below, along with the time taken to walk from building to building.



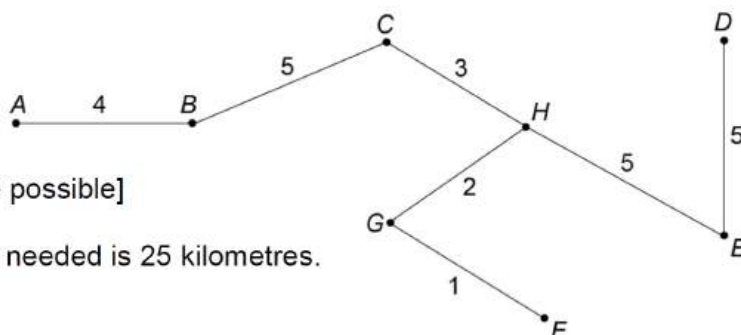
- How many nodes in the graph have an even degree?
4 – S, R, U, W
- Calculate the fastest route to get from building P to building V. **PQRV – 18 minutes**
- A university student wants to put up pamphlets at each building. Write down a path she could take, without walking along any particular road more than once.
PQSTRVUW or TPSQRVUW or VUWRTSQP or VRWUTSQP or QSPTRVUW or VUWRSQPT
(many possible solutions)
- Draw a minimum spanning tree for this network.



- 10 The network diagram shows the tracks connecting 8 picnic sites in a nature park. The vertices A to H represent the picnic sites. The weights on the edges represent the distances along the tracks between the picnic sites, in kilometres.



- Each picnic site needs to provide drinking water. The main water source is at site A. By drawing a minimum spanning tree in the space below, calculate the minimum length of water pipes required to supply water to all the sites if the water pipes can only be laid along the tracks.



[Other minimum spanning trees are possible]

The minimum length of water pipes needed is 25 kilometres.

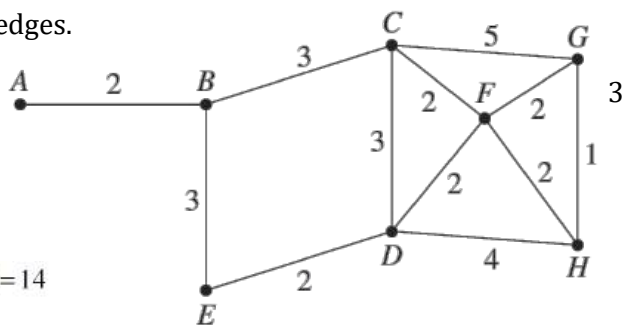
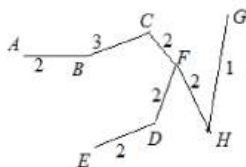
- One day, the track between C and H is closed. State the vertices that identify the shortest path from C to E that avoids the closed track.

Path goes from C to G to H to E.

11 The diagram represents a network with weighted edges.

- a. Draw a minimum spanning tree for this network in the space below and determine its length.

$$\text{Minimum length} = 2 + 3 + 2 + 2 + 1 + 2 + 2 = 14$$



- b. The network is revised by adding another vertex, K. Edges AK and CK have weights of 12 and 10 respectively, as shown. What is the length of the minimum spanning tree for this revised network?

$$\text{Minimum length} = 2 + 3 + 2 + 2 + 1 + 2 + 2 + 10 = 24$$

